

Robot Trajectory from Dataset

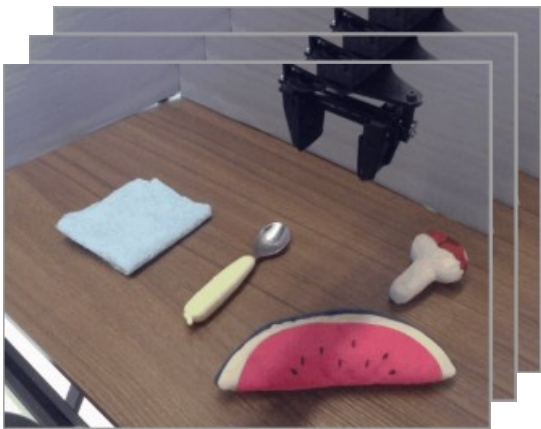
Task Instruction

Put the watermelon on the towel

State Info

Proprio 1: $[\Delta x, \Delta \theta, \Delta \text{Grip}]$

Observations



1. Describe scene

Prismatic-VLM

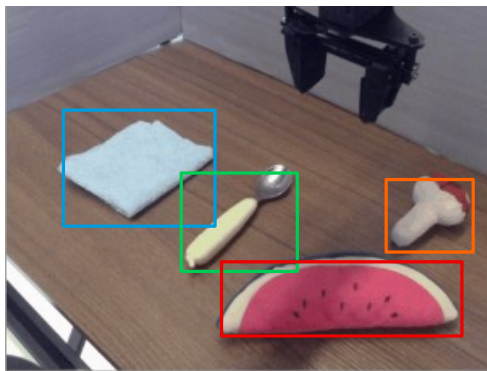


Please describe the scene.

*This scene has a **watermelon**, **spoon**, **towel**, and **mushroom**.*

2. Extract bounding boxes

Grounding DINO



3. Compute motion primitives

Proprio $\rightarrow$ Primitives

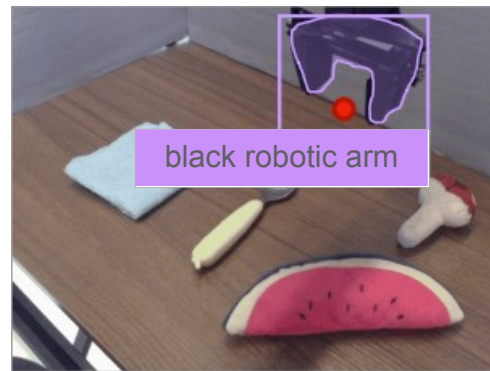


Proprio 1: $[\Delta x, \Delta \theta, \Delta \text{Grip}]$

{right, down, down, ...}

4. Compute gripper position

OWL + SAM



5. Generate plans + subtasks

Gemini LLM



Explain the plan, subtask, and movement for each step, given:

Task

Desc.

Moves

Plan: Go to watermelon, grasp, go to towel, release

Subtasks:

1. The robot needs to grasp the watermelon, so first must move to it
2. The robot is at the watermelon, so it can now grasp it
3. ...

Moves:

1. The watermelon is right of the robot, so it must move <right>
2. The watermelon is below the robot, so it must move <down>
3. ...